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Effect of landfast sea ice on coastal currents driven by the wind

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Abstract. The present study examines the effects of landfast sea ice on the response of a homogeneous coastal ocean to wind forcings under the longwave approximation. For a fluctuating wind, continental shelf waves are induced by an infinite stress curl due to the existense of the ice edge as well as by the torque due to the topographic change. The amplitude and sign of each shelf wave mode excited by the forcing depend strongly on the ice edge location. The alongshore velocity field, which can be represented as the sum of all the shelf wave modes, has a maximum in amplitude at the ice edge with a strong lateral shear, which is a result of the infinite stress curl. The amplitudes of alongshore velocity and sea level near the coast are greatly reduced as the ice edge advances. The response to a steady wind is also examined, and it is found that even for the same wind, the velocity distributions depend strongly on the ice edge location. For instance, the current direction near the coast is the same as that of the wind stress in the case of no fast ice, whereas it is opposite in the case where the ice edge is located at the shelf break. The present theory is applied to the continental shelf off Enderby Land, Antarctica, and it gives the possible explanation for the observed features that the current variability, its coherence with the wind, and the prevailing current direction depend much on the extent of landfast sea ice.

1. Introduction

In polar regions some coastal oceans are occasionally covered by landfast sea ice (sea ice fixed to a landmass). For instance, a part of the coastal area in the Antarctic Continent is surrounded by landfast sea ice. Since the stationary ice prevents the water beneath it from feeling the wind blowing over it, the response of the ocean to the wind strongly depends on the existence of the fast ice. Even for a spatially uniform wind, at the fast ice edge a large stress curl acts on the ocean, which gives rise to excitation of the currents [*Clarke*, 1978].

Observationally, *Ohshima et al.* [this issue] suggested that current variability under landfast sea ice in Lützow-Holm Bay off Enderby Land, Antarctica, depends on the extent of the landfast ice. They further inferred that the excitation of the first-mode shelf wave is greatly reduced as the landfast ice advances. In Hudson Bay, *Lepage and Ingram* [1991] showed that low-frequency motion, driven by the wind, is considerably reduced by the presence of the landfast ice.

The response of the coastal ocean to the fluctuating wind is interpreted by the dynamics of the coastally trapped waves (CTWs), and much theoretical and observational effort has been made for CTWs [*Mysak*, 1980]. The response of the coastal ocean to the steady wind can be interpreted by arrested topographic waves (ATWs) [*Csanady*, 1978].

The purpose of this study is to examine the response of the coastal ocean to fluctuating and steady winds in the presence of landfast sea ice in analytical models. The models are also applied to a specific case; the case of the continental shelf off Enderby Land, Antarctica.

2. Solution for Periodic Forcing (Shelf Wave Response)

2.1. Formulation and General Solution

The model equations are the same as those considered by *Gill and Schumann* [1974] except for the inclusion of landfast sea ice. A right-handed coordinate system is used with the positive x axis pointing to the offshore direction and the y axis lying along the straight coastline at $x = 0$ (Figure 1). The water depth h varies only in the cross-shelf direction, i.e., $h = h(x)$. For simplicity we consider the case where the fast ice edge is infinitely located at $x = m$, parallel to the coast (Figure 1).

For linear, nondivergent motions under the inviscid and longwave approximations the vertically integrated shallow water equations have the form

$$-fv = -g\frac{\partial \eta}{\partial x} \quad (1)$$

$$\frac{\partial v}{\partial t} + fu = -g\frac{\partial \eta}{\partial y} + \frac{Y}{h}, \quad (2)$$

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where u and v are the depth-averaged components of velocity in the x and y directions, respectively; η is the elevation of the sea surface above the equilibrium level; Y is the y component of wind stress acting on the ocean surface, divided by the density of the water; g is the acceleration of gravity; and f is the Coriolis parameter, assumed to be constant. Consider the case where the wind is uniform in the cross-shelf direction. Since the wind stress cannot be applied to the ocean underneath the fast ice, Y is represented by

$$Y = Y_0 H(x-m), \tag{3}$$

where $H(x)$ is step function. The relevant boundary conditions are

$$\begin{cases} u = 0 & x = 0 \\ v = 0 & x = l. \end{cases} \tag{4}$$

By introducing a stream function ψ such that $hu = -\partial\psi/\partial y$ and $hv = \partial\psi/\partial x$, (1) and (2) can be written as

$$\frac{\partial^2}{\partial x \partial t}\left(\frac{1}{h}\frac{\partial\psi}{\partial x}\right) + \frac{f}{h^2}\frac{dh}{dx}\frac{\partial\psi}{\partial y} = -\frac{1}{h^2}\frac{dh}{dx}Y_0 H(x-m) + \frac{1}{h}Y_0\delta(x-m). \tag{5}$$

Physically, the vertically integrated form of (5) represents the sum of the transports in the surface Ekman layer and the layer below [*Gill and Schumann*, 1974]. When we consider the motion below the Ekman layer on the assumption that the Ekman layer is sufficiently thin compared with the water depth, the v fields are the same as those derived from the vertically integrated form of (5), although the u fields must be corrected for the Ekman layer component in the forced region [*Gill and Schumann*, 1974; *Middleton*, 1988]. Thus we will treat this vertically integrated form of (5) and consider mainly the v fields hereinafter.

Here we briefly explain (5), the forced vorticity equation, in physical terms. The vorticity input is primarily caused by the response of the surface Ekman layer to the wind stress. The second term on the right-hand side of (5) is the term that is newly introduced because of the presence of the landfast ice. It represents the vertical motion due to divergence/convergence in the Ekman layer. For the spatially uniform wind it occurs at the ice edge where there is an infinite stress curl. The first term on the right-hand side of (5) represents the vorticity input caused by the topographic change. We call this term "topographic torque" hereinafter. As in the discussion of *Gill and Schumann* [1974, section 8], this term arises from the condition of no flux at the coast with the nondivergent motion: the motion below the Ekman layer is steered by shrinking or stretching due to the compensated flow of the normal Ekman flux on the offshore side of the ice edge.

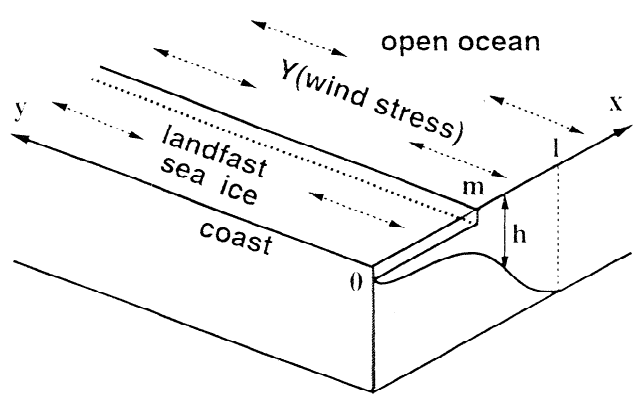

Figure 1. Schematic configuration of the analytical model.

The homogeneous problem, (right-hand side of (5))=0, can be solved by assuming

$$\psi(x,y,t) = \sum F_n(x)\phi_n(y,t). \tag{6}$$

Then the cross-shelf eigenfunctions $F_n(x)$ satisfy

$$\frac{d}{dx}\left(\frac{1}{h}\frac{dF_n}{dx}\right) - \frac{f}{c_n}\frac{F_n}{h^2}\frac{dh}{dx} = 0, \tag{7}$$

where c_n is the separation constant with the boundary conditions, $F_n(0) = 0$, and $dF_n/dx = 0$ at $x = l$, where the orthonormality condition is

$$\int_0^l \frac{1}{h^2}\frac{dh}{dx}F_n F_m dx = \delta_{nm}. \tag{8}$$

The forcing terms (right-hand side of (5)) are now expanded in terms of $F_n(x)$:

$$-Y_0 H(x-m) = \sum F_n(x)\beta_n(y) \tag{9}$$

$$Y_0\frac{h}{dh/dx}\delta(x-m) = \sum F_n(x)\alpha_n(y). \tag{10}$$

Applying the operator $\int_0^l \frac{1}{h^2}\frac{dh}{dx}F_n dx$ to (9) with (8) gives

$$\beta_n(y) = Y_0 b_n \tag{11}$$

$$b_n = -\int_0^l H(x-m)\frac{1}{h^2}\frac{dh}{dx}F_n dx = -\int_m^l \frac{1}{h^2}\frac{dh}{dx}F_n dx. \tag{12}$$

Applying the operator $\int_0^l \frac{1}{h^2}\frac{dh}{dx}F_n dx$ to (10) with (8) gives

$$\alpha_n(y) = Y_0 d_n \tag{13}$$

$$d_n = \int_0^l \frac{\delta(x-m)}{h}F_n dx = \frac{F_n(m)}{h(m)}. \tag{14}$$

Substitution of (6), (9), and (10) into (5) with (11) and (13) gives

$$\frac{1}{c_n}\frac{\partial\phi_n}{\partial t} + \frac{\partial\phi_n}{\partial y} = \frac{Y_0}{f}(b_n + d_n). \tag{15}$$

The sum of b_n and d_n represents the contribution of each mode excited by the forcing. Integrating (15) obtains

$$\phi_n(y,t) = \phi_n(0, t - \frac{y}{c_n}) + \frac{b_n + d_n}{f}\int_0^y Y_0(\xi, t + \frac{\xi - y}{c_n})d\xi. \tag{16}$$

The real alongshore velocity, represented as the sum of all the modes, becomes

$$v(x,y,t) = \sum \phi_n h^{-1} \frac{dF_n}{dx}. \tag{17}$$

The sea level change across the shelf is given by

$$\eta(x,y,t) = \frac{f}{g} \sum \phi_n \left[\frac{F_n(x)}{h(x)} - \frac{F_n(l)}{h(l)} + \int_l^x \frac{1}{h^2} \frac{dh}{dx} F_n(x) dx \right]. \tag{18}$$

2.2. Solution for the Exponential Shelf Profile

The coefficients b_n and d_n as functions of the ice edge location m can be obtained analytically for the case of the exponential shelf profile:

$$\begin{cases} h(x) = h_0 e^{2\lambda x} & 0 < x < l \\ h(x) = h_0 e^{2\lambda l} & x > l. \end{cases} \tag{19}$$

In this case, $F_n(x)$ and c_n can be given by $F_n(x) = A_n e^{\lambda x} \sin \beta_n x$ and $c_n = -2\lambda f(\beta_n{}^2 + \lambda^2)^{-1}$, where $\lambda \sin \beta_n l + \beta_n \cos \beta_n l = 0$ and $A_n{}^2 = h_0 \lambda^{-1}[l + \lambda/(\beta_n{}^2 + \lambda^2)]^{-1}$ [*Gill and Schumann*, 1974]. Then the coefficients b_n and d_n can be calculated from (12) and (14):

$$b_n = -2\frac{A_n}{h_0} \frac{\lambda}{\lambda^2 + \beta_n{}^2} e^{-\lambda m} (\lambda \sin \beta_n m + \beta_n \cos \beta_n m) \tag{20}$$

$$d_n = \frac{A_n}{h_0} e^{-\lambda m} \sin \beta_n m. \tag{21}$$

Figure 2 shows the coefficients b_n and d_n for the first mode (i.e., $n = 1$) as a function of the ice edge location m. For the case where the wind direction is toward the positive y direction the first term of the right-hand side of (5), which gives the coefficients b_n, produces the negative vorticity in the offshore side of the ice edge, whereas the second term of the right-hand side of (5),

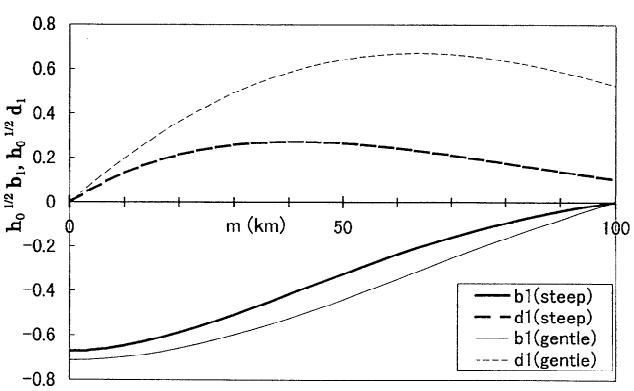

Figure 2. The coefficients $h_0{}^{1/2} b_1$ and $h_0{}^{1/2} d_1$ as functions of the ice edge location m in the cases of the exponential shelf profiles with $f = -1.34 \times 10^{-4}$ S^{-1} (corresponding to latitude 67°S) and $l = 100$ km. The steep shelf case (solid lines) has $h_0 = 100$ m and $\lambda l = 1.7$ (maximum depth is ~4000 m), and the gentle shelf case (dotted lines) has $h_0 = 500$ m and $\lambda l = 0.69$ (maximum depth is ~2000 m).

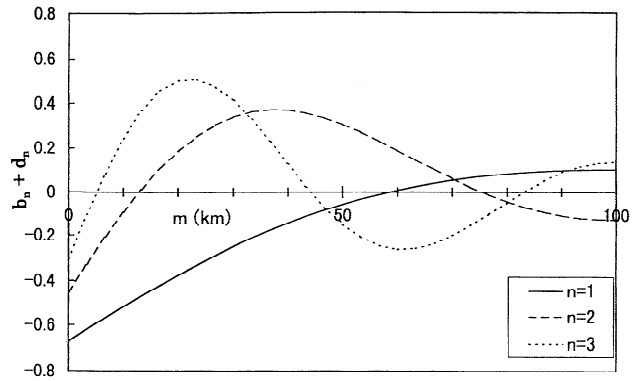

Figure 3. The coefficients $(b_n + d_n)$ for n =1, 2, and 3 as functions of the ice edge location m in the case of the steep exponential shelf ($h_0 = 100$ m, $\lambda l = 1.7$, and $l = 100$ km).

which gives the coefficient d_n, produces the infinite positive vorticity at the ice edge. Therefore, for the first mode, which has no node over the shelf, b_1 and d_1 have an opposite sign, as shown in Figure 2. The value of d_1, which is zero for $m = 0$, increases with the offshore extent of the ice edge location and finally exceeds the absolute value of b_1. As shown in Figure 2, the term d_1 is relatively more important in the gentle slope case than in the steep slope case because the topographic torque, which gives the term b_1, becomes relatively smaller in the gentle slope case.

The form of (21) tells us that d_n changes its sign by n times from $m = 0$ to $m = l$ since β_n is a kind of cross-shore wavenumber. Figure 3 shows $(b_n + d_n)$ for each mode as a function of the ice edge location m. It demonstrates that the amplitude and sign of each mode excited by the forcing strongly depend on ice edge location. The dominant mode changes from the first mode in the case of no fast ice ($m = 0$ km) to the third mode in the case of $m = 20$ km and then to the second mode in the case of $m = 40$ km.

2.3. Application to the Continental Shelf off Enderby Land

The system of equations (5)–(18) can be solved numerically for an arbitrary depth profile. Here the theory is applied to the continental shelf off Enderby Land, Antarctica, where *Ohshima et al.* [this issue] suggested that the ocean response to the wind forcing strongly depends on the extent of the landfast sea ice over this shelf. Figures 4a and 4b show the cross-shelf structures of the alongshore velocity v_n and the sea level deviation η_n, respectively, for the first- , second- , and third-mode shelf waves for a typical depth profile off Enderby Land (Figure 4d). Figure 4c shows the coefficients $(b_n + d_n)$ for the first, second, and third modes as a function of the ice edge location m. The validity of the numerical method of solutions is assured by comparison with the analytic solution of the exponential profile case. It is found from Figure 4c that the amplitude and sign of each mode strongly depend on the ice edge location, as

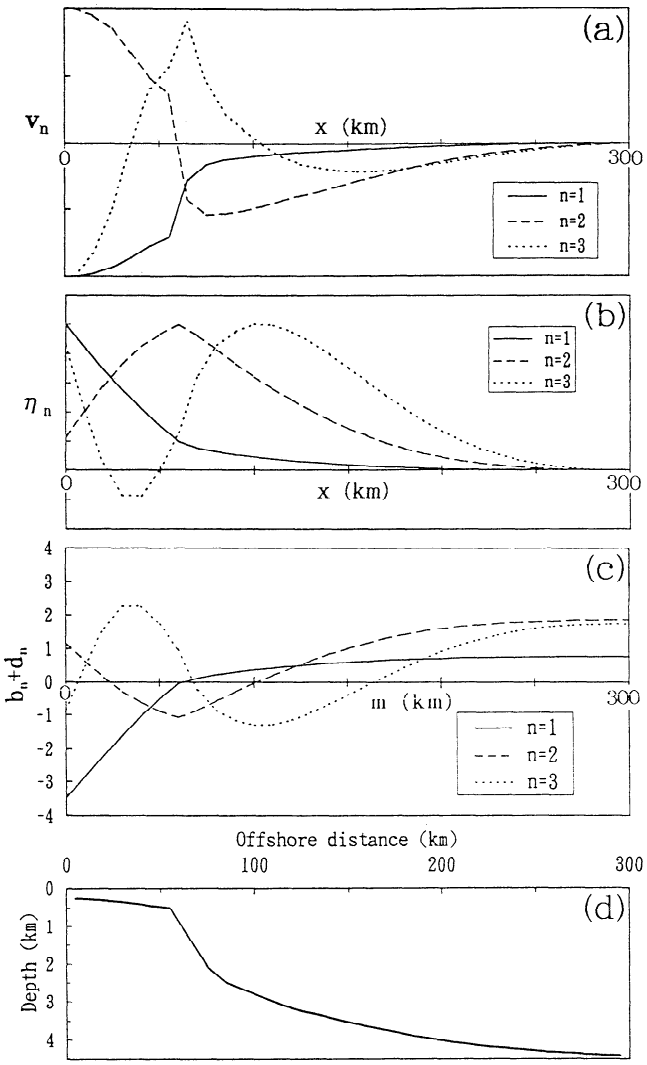

Figure 4. Amplitudes of (a) the alongshore velocity v_n and (b) the sea level deviation η_n in the cross-shelf direction for the first- , second- , and third-mode shelf waves for (d) a typical depth profile off Enderby Land, Antarctica. (c) The coefficients $(b_n + d_n)$ for the first, second, and third modes as a function of the ice edge location m.

in the case of the exponential depth profile. The first mode is the dominant mode in the case of no fast ice, while it is much less excited as the ice edge advances toward the shelf break.

In what follows we consider the ocean response to an oscillatory wind stress of the form

$$Y = \begin{cases} 0 & y < 0 \\ Y_0 \sin \omega t & 0 < y < d \\ 0 & y > d, \end{cases} \tag{22}$$

where ω is the frequency. This form of the wind stress variation is representative of the first approximation for the Antarctic coastal ocean [*Ohshima et al.*, this issue]. The alongshore velocity can be represented by the sum of all the shelf wave modes. From (16), (17), and (22), $v(x, y_0, t_0)$ for $0 < y_0 < d$ is given by

$$v(x, y_0, t_0) = \sum \frac{2c_n(b_n+d_n)Y_0}{\omega f} \sin \frac{\omega y_0}{2c_n} \sin \omega(t_0 - \frac{y_0}{2c_n}) h^{-1} \frac{dF_n}{dx}. \tag{23}$$

For the continental shelf off Enderby Land we take f=-1.34 s^{-1}, corresponding to 67°S, and the wind stress is assumed to have an amplitude of 0.1 N m^{-2} for an 8 day period [*Ohshima et al.*, this issue]. Figure 5 shows an instantaneous field of the alongshore velocity calculated from (23) for the case where the ice edge is located at the shelf break (m = 60 km). The positive value represents westward velocity for the case of the shelf off Enderby Land. We find that the maximum of the velocity amplitude occurs at the ice edge with a strong shear, which is a result of the infinite vorticity input at the ice edge. In this specific case the second mode is dominant (Figure 4c).

To demonstrate the difference from the case of no fast ice, $x - t$ diagrams of the alongshore velocity at y_0 = 1500 km are shown for the cases of m = 0 km (Figure 6a) and m = 60 km (Figure 6b). The velocity near the coast in the case of m = 60 km is particularly damped, less than one tenth of that of the no fast ice case, mainly because of slight excitation of the first mode (see Figure 4c). In the case of no fast ice the velocity field is mainly determined by the first-mode shelf wave. Hence the phase lag of the velocity to the wind at y_0 = 1500 km is ~1.5days, very close to the value predicted by the first-mode shelf wave, $y_0/2c_1$ = 1.5

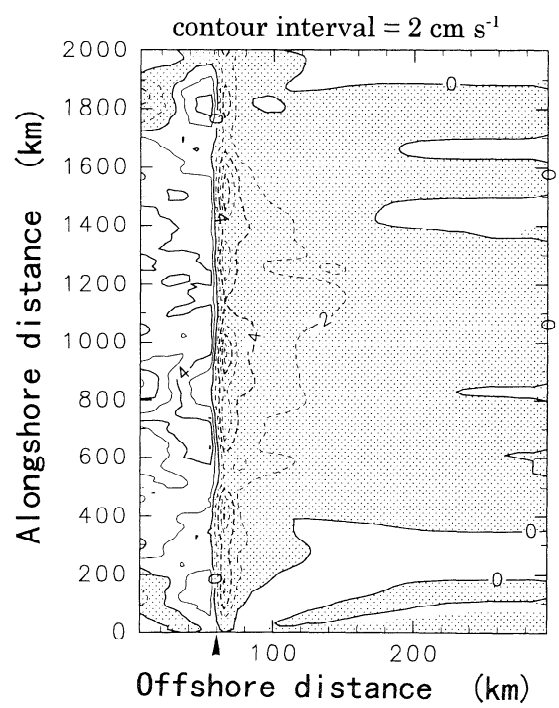

Figure 5. An instantaneous field of the alongshore velocity calculated from (23) for the case where the ice edge is located at the shelf break (m = 60 km). The arrow indicates the location of the ice edge. The amplitude of the wind stress is 0.1 N m^{-2} for an 8 day period. The shading and dashed contours indicate negative velocity.

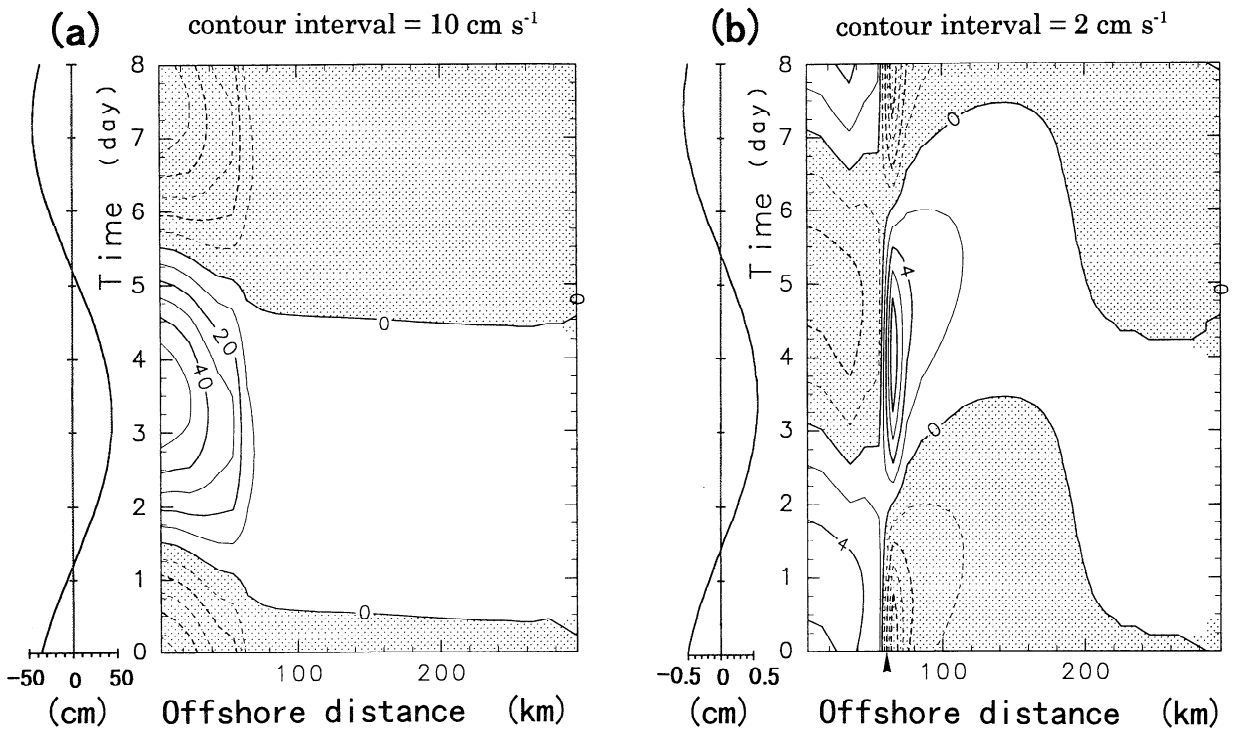

Figure 6. The $x-t$ diagrams of the alongshore velocity at $y_0 = 1500$ km for (a) the case of $m = 0$ km and (b) that of $m = 60$ km. The amplitude of the wind stress is 0.1 N m^{-2} for an 8 day period for both cases. The sea level variations at the coast are also shown. Note that the contour intervals and the sea level scales are different between Figures 6a and 6b. The shading and dashed contours indicate negative velocity. The arrow in Figure 6b indicates the location of the ice edge.

days, where $c_1 (= 600$ km d$^{-1})$ is the phase speed of the first-mode. The sea level variations at the coast are also shown in Figures 6a and 6b. The sea level amplitude is reduced by 2 orders of magnitude by the presence of the fast ice and because of slight excitation of the first mode. The results of Figure 6 are consistent with the observations off Enderby Land [*Ohshima et al.*, this issue]. The alongshore velocity and sea level near the coast are coherent with the wind with a lag time of 1.5 days when the fast sea ice is nearly absent (corresponding to the case of $m = 0$ km), but they do not show significant coherence when the fast ice margin is stably located at the shelf break (corresponding to the case of $m = 60$ km).

As shown in Figures 5 and 6b, strong current shear occurs at the ice edge. To examine the shear fields for the wind forcing in the presence of landfast ice, the cross-shelf distribution of the shear intensity is calculated as a function of the ice edge location (Figure 7), where the shear intensity is defined as $\overline{|\partial v/\partial x|}$ (the bar means average over one period of the wind forcing). It is found from Figure 7 that as long as the ice edge is located over the shelf, relatively strong shear occurs at the ice edge (trace the dotted line) and at the shelf break. The strong lateral shear at the ice edge originates from an infinite stress curl at the ice edge (the second term on the right-hand side of (5)). In contrast, the strong shear at the shelf break originates from the topographic torque with the large depth change (the first term on the right-hand side of (5)). For the case where the ice edge is located off the shelf break, strong shear does not occur because both the first and second terms become small.

3. Solution for Steady Forcing

Over the Antarctic coast, for instance, the prevailing wind direction is generally westward throughout the year. In this section we examine the response to the steady wind forcing in the presence of landfast sea ice. We consider a similar situation to that of Figure 1 under the condition that the uniform alongshore wind stress Y_0 is applied to the region $0 < y < d$ with the bottom friction included. Under linear, nondivergent, and longwave approximation, in the shelf region,

$$-fv = -g\frac{\partial \eta}{\partial x}, \tag{24}$$

$$fu = -g\frac{\partial \eta}{\partial y} - \frac{r}{h}v + \frac{Y_0 H(x-m)}{h}, \tag{25}$$

where r is the coefficient of the bottom friction. In comparison with the periodic forcing case the term $\partial v/\partial t$ is replaced by rv/h. This steady problem can be reduced to a heat equation forced by a delta function by solving with respect to sea surface deviation η [e.g., *Csanady*,

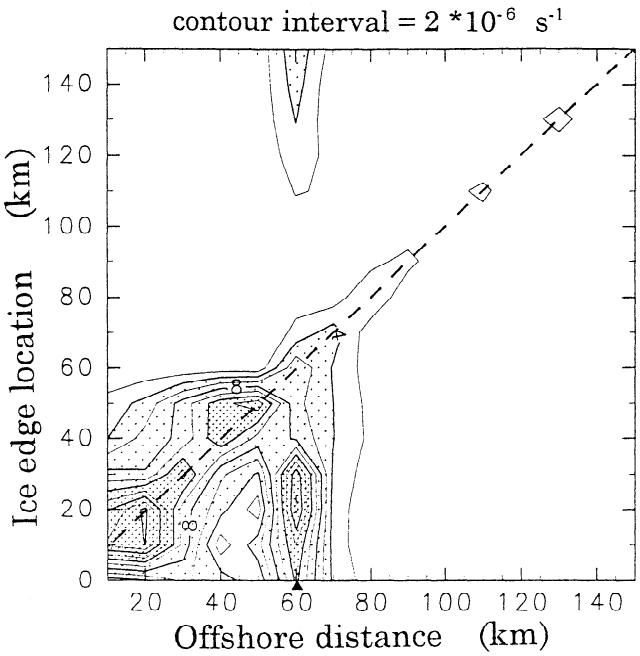

Figure 7. The cross-shelf distribution (abscissa) of the current shear intensity as a function of ice edge location (ordinate), where the situation and parameters are similar to those in Figure 6. The shear intensity is defined as $\overline{|\partial v/\partial x|}$, where the bar means the average over one period of the wind forcing. The shading indicates high shear intensity. The triangle indicates the location of the shelf break. The dashed line corresponds to the location of the ice edge.

1978; *Chapman et al.*, 1986]. Also, the problem can be regarded as the "arrested wave" version of the periodic solution described in section 2. In this paper we solve the problem by analogy to the periodic solution.

By introducing a stream function ψ the counterpart of (5) becomes

$$r\frac{\partial}{\partial x}\left(\frac{1}{h^2}\frac{\partial\psi}{\partial x}\right)+\frac{f}{h^2}\frac{dh}{dx}\frac{\partial\psi}{\partial y} = -\frac{1}{h^2}\frac{dh}{dx}Y_0H(x-m)+\frac{1}{h}Y_0\delta(x-m). \quad (26)$$

We assume that

$$\psi(x,y)=\sum F_n(x)\phi_n(y). \quad (27)$$

$F_n(x)$ satisfies

$$r\frac{d}{dx}\left(\frac{1}{h^2}\frac{dF_n}{dx}\right)-\frac{f}{c_n}\frac{F_n}{h^2}\frac{dh}{dx}=0, \quad (28)$$

where c_n is the separation constant. Equation (26) can be solved by the forcing terms being expanded in terms of $F_n(x)$, similar to the periodic forcing case (5). The counterpart of (15) is

$$\frac{1}{c_n}\phi_n+\frac{d\phi_n}{dy}=\frac{Y_0}{f}(b_n+d_n). \quad (29)$$

Equation (29) can be integrated to obtain

$$\phi_n(y)=\begin{cases}0 & y<0\\ \frac{Y_0}{f}(b_n+d_n)c_n(1-e^{-y/c_n}) & 0<y<d\\ \frac{Y_0}{f}(b_n+d_n)c_n(e^{d/c_n}-1)e^{-y/c_n} & y>d.\end{cases} \quad (30)$$

The alongshore velocity at $y=y_0(0<y_0<d)$ is given by

$$v(x,y_0)=\frac{Y_0}{f}\sum(b_n+d_n)c_n(1-e^{-y_0/c_n})h^{-1}\frac{dF_n}{dx}. \quad (31)$$

The sea level deviation at $y=y_0$ is given by

$$\eta(x,y_0)=\frac{Y_0}{g}\sum\left\{(b_n+d_n)c_n(1-e^{-y_0/c_n})\cdot\left[\frac{F_n(x)}{h(x)}-\frac{F_n(l)}{h(l)}+\int_l^x\frac{1}{h^2}\frac{dh}{dx}F_n(x)dx\right]\right\}. \quad (32)$$

As a specific case, we also adopt the bathymetry off Enderby Land (Figure 8d) as in section 2.3. Figure 8a shows the cross-shelf structure of the alongshore velocity v at $y_0=1500$ km calculated from (31) for the case where the ice edge is located at the shelf break ($m=60$ km), the wind stress is 0.15 N m^{-2}, and $r=5.0\times10^{-4}$ m s^{-1}. The value of r is based on *Chapman et al.* [1986] but is assumed to be constant for simplicity. The velocity v can be divided into two components, i.e., $v=v_b+v_d$, where v_b and v_d are the terms multiplied by b_n and d_n in (31), respectively. The term v_b originates from the first term on the right-hand side of (26) and the term v_d originates from the second term on the right-hand side of (26). The structures of v_b and v_d are also shown in Figure 8a. In this case, v_b gives negative vorticity for $x>m$ by the topographic torque, while v_d gives strong positive vorticity at $x=m$ by the effect of the infinite stress curl.

Figure 8b gives a comparison of the structure of the alongshore velocity for the cases of $m=60$ km and $m=0$ km (no fast ice) for the same wind forcing of 0.15 N m^{-2}. For the case of no fast ice the structure of v, having the only v_b component, gives negative vorticity over the whole region. The structure of v is quite different between the two cases. Near the coast the current direction is the same as the wind stress direction in the case of no fast ice, while the current direction is opposite the wind stress direction in the case of $m=60$ km. Figure 8c gives a comparison of the sea level structure between the cases of $m=60$ km and $m=0$ km. At the coast, sea level is higher by ~7 cm in the case of no fast ice.

According to the extension of the theory of *Csanady* [1978] the alongshore volume transport of the current is determined precisely by the integration of the onshore Ekman transport at the ice edge over the eastward coast. The volume transport VT at $y=y_0$ for a wind stress $Y(x,y)$ is given by

$$VT=-\int_{-\infty}^{y_0}\frac{Y|_{x=m}}{f}dy. \quad (33)$$

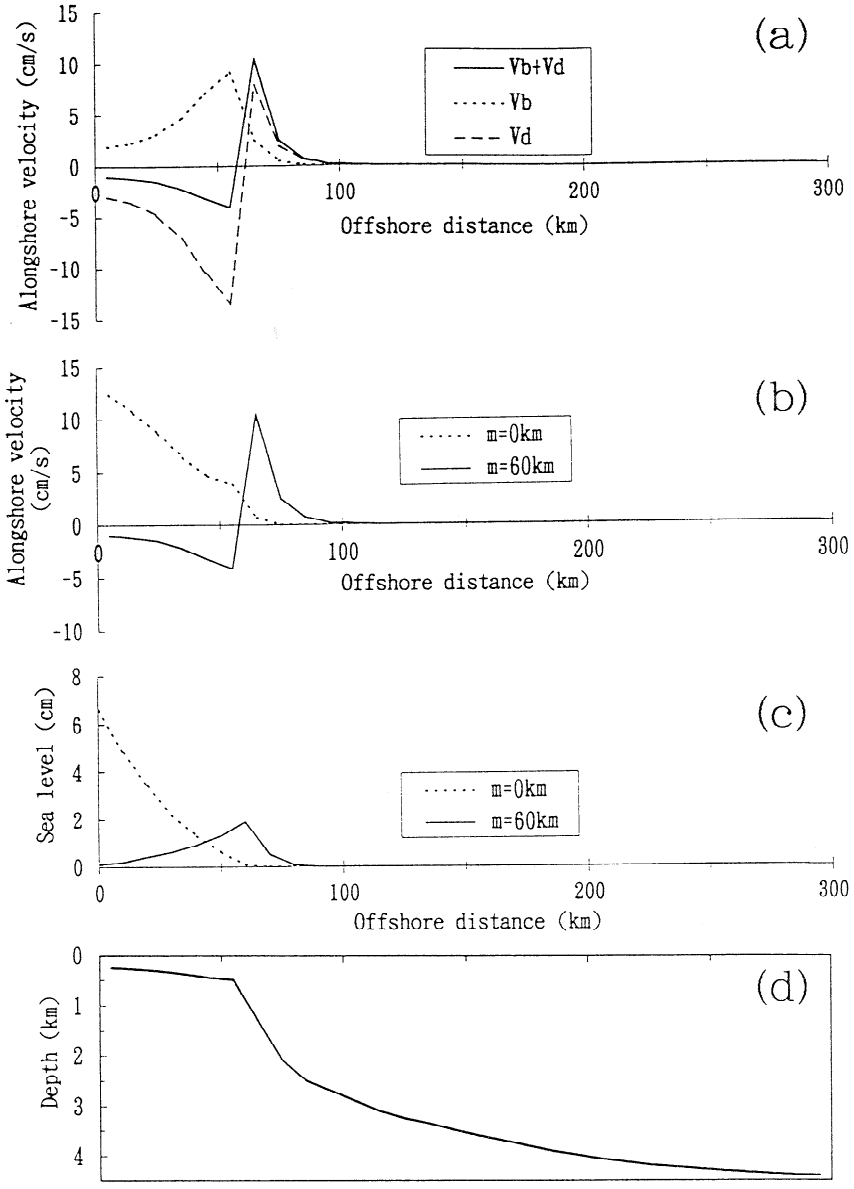

Figure 8. (a) The cross-shelf structure of the alongshore velocity v and its two components v_b and v_d calculated from (31) at $y_0 = 1500$ km for the case where the ice edge is located at the shelf break ($m = 60$ km), the wind stress is 0.15 N m^{-2}, and $r = 5.0 \times 10^{-4}$ m s^{-1}. (b) The cross-shelf structure of the alongshore velocity for the cases of $m = 60$ km and $m = 0$ km (no fast ice) for the same wind forcing of 0.15 N m^{-2} and for $r = 5.0 \times 10^{-4}$ m s^{-1}. (c) As in Figure 8b but for the sea level structure, where the sea level at the offshore end is referred to zero level. (d) The typical depth profile off Enderby Land used in the calculation.

For a uniform wind stress in the cross-shore direction the volume transport becomes the same, independent of ice edge location and bottom friction, even for the case where the current structures are quite different, such as in Figure 8b. From (33), VT for the case of Figure 8b is reduced to $Y_0 y_0/f (= 1.64 \times 10^6$ m^{3}s$^{-1})$.

When we consider a variation whose timescale is much longer than the spin down time (h/r), in which the term rv/h is much greater than the term $\partial v/\partial t$ in the momentum equation, the steady formulation is a better approximation than the inviscid periodic formulation. In the case of the continental shelf off Enderby Land, taking $h = 500$ m and $r = 5.0 \times 10^{-4}$ m s^{-1}, the spin down time is estimated to be ~12 days. As shown by *Ohshima et al.* [this issue], off Enderby Land the following features are observed for a longer period variation. The current direction near the coast is the same as the wind stress direction when the fast ice is nearly absent while it is opposite after the fast ice is stably located at the shelf break. The adjusted sea level averaged over the former period is higher by ~10–15 cm than that in the latter period. These results are consistent with the results of Figures 8b and 8c, although only qualitatively.

4. Summary and Discussion

The present study examines the effects of landfast sea ice on the response of a homogeneous coastal ocean to wind forcings under the longwave approximation. The response to a fluctuating wind can be described by continental shelf waves. Shelf waves are induced by an infinite stress curl at the ice edge as well as by the topographic torque (right-hand side of (5)). The amplitude and sign of each mode excited by the forcing depends strongly on the ice edge location. For an exponential shelf profile the dominant mode changes from the first mode to the third mode and then to the second mode as the ice edge advances (Figure 3). The alongshore velocity field, which can be represented as the sum of all the shelf wave modes, has its maximum amplitude at the ice edge with a strong shear (Figures 5 and 7), which is a result of the infinite stress curl. The amplitudes of alongshore velocity and sea level near the coast are greatly reduced as the ice edge advances (Figures 6a and 6b). The response to a steady wind can be interpreted by the "arrested" version of shelf wave response, and it is found that even for the same wind, the velocity distributions depend strongly on the ice edge location. For instance, the current direction near the coast is the same as that of the wind stress in the case of no fast ice, whereas it is opposite in the case where the ice edge is located at the shelf break (Figure 8b).

The present study assumes an homogeneous ocean and alongshore uniformity. Although these two assumptions are idealistic, they seem to be good approximations as the first order for an Antarctic coastal ocean, where the bottom topography is relatively uniform along the continent and the stratification parameter, $S = Na/f$, is generally < 1 (N is Brunt-Väisälä frequency and a is bottom slope). Indeed, the present theory is applied to the continental shelf off Enderby Land, Antarctica, and it gives a possible explanation for some of the observed features. The periodic solution is consistent with the fact that the current variability, which may be associated with the first-mode shelf wave, is greatly damped with the advancement of the landfast sea ice. The steady solution is consistent with the change in the prevailing current direction near the coast in accordance with the advancement of the fast ice.

Here we briefly discuss the effects of density stratification. *Clarke* [1978] gave the basic theory to describe the wind-driven motion near landfast sea ice in the presence of density stratification. He showed, using a two-layer model with no coastal boundary, that an infinite stress curl at the edge of landfast sea ice gives rise to upwelling/downwelling and the corresponding currents along the ice edge and that these motions depend only on the local forcing. In the realistic ocean, however, the bottom topography and the coastal boundary cause the steered motions to propagate along the coast with the coast to the left (right) in the Southern (Northern) Hemisphere. Thus the motion cannot be determined only by the local forcing. For a realistic case where both bottom topography and density stratification are included the response of the coastal ocean can be described by the CTWs, which have hybrid properties of internal Kelvin waves and barotropic shelf waves. *Lopez-Mariscal and Clarke* [1993] examined the influence of wind stress curl on low-frequency motions over the continental shelf. They showed that the dominant vertical shear with the tilting of isopycnals is associated with nonzero wind stress curl (the surface Ekman convergence/divergence). Inferring from their study, at the ice edge, where the Ekman convergence/divergence is very strong, the vertical shear and tilting of isopycnals are expected to be particularly dominant under the inclusion of density stratification.

In polar coastal oceans, unlike the case of the continental shelf off Enderby Land, the alongshore uniformity of the bottom topography and the ice edge location is not necessarily satisfied. Even for such cases, the large stress curl that occurs at the ice edge gives rise to significant motions with strong shears, and then the motions, trapped by the coastal bottom topography, propagate in the direction with the coast to the left (right) in the Southern (Northern) Hemisphere. Plausibly, this process is an important factor for determining the flow field in polar coastal oceans.

The present study examines how the landfast ice affects the current field. Inversely, the current field might affect the edge location of landfast ice. For instance, the strong current shear might cause shear cracks in the landfast sea ice. In the Antarctic coast the extent of the landfast sea ice is usually confined to the shelf break [*Ishikawa et al.*, 1996; *Ohshima et al.*, this issue], which is a dynamically important site where the current shear tends to be strong (Figure 7). The existence of the ice edge also induces the strong current shear (Figures 5 and 7). The dynamical interaction between current and landfast ice would seem to be an interesting theme for future study.

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